Configure IPv6 Addressing

Task 1 : Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Operating System: Windows 11

Network Adapter: Wi-Fi

Command Used:

netsh interface ipv6 add address "Wi-Fi" fd12:3456:789a::10

Verification Command:

ipconfig

Static IPv6 Address Assigned: fd12:3456:789a::10

Result: The static IPv6 address was successfully added to the Wi-Fi network adapter. The ipconfig command displayed the newly assigned IPv6 address along with the existing IPv6 and IPv4 configuration.

Conclusion: IPv6 addressing was configured and verified successfully using Windows Command Prompt.

Task 2 : Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

To configure a Unique Local IPv6 Address on Windows and verify its configuration.

Procedure

1. Open PowerShell as Administrator.
2. Check the IPv6 addresses using:

Get-NetIPAddress -AddressFamily IPv6

3. The Wi-Fi adapter already had the following ULA configured:
    - IPv6 Address: fd12:3456:789a::10
    - Prefix Length: 64
    - Interface: Wi-Fi
4. Verify the configuration using ipconfig.
5. To test connectivity, another LAN device with a ULA is required. Since a second device was unavailable, the ping test could not be performed.

Result

The ULA fd12:3456:789a::10/64 was verified on the Wi-Fi adapter.

Conclusion

ULA addresses are used for private communication within local networks. The address configuration was verified, but connectivity testing remains pending.

Task 3 : Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

To observe temporary IPv6 address behavior after reconnecting to the network.

Procedure

1. Open PowerShell and run Get-NetIPAddress -AddressFamily IPv6.
2. Run ipconfig /all to inspect the IPv6 configuration of the Wi-Fi adapter.
3. Record the IPv6 addresses displayed before reconnecting.
4. Disconnect and reconnect Wi-Fi.
5. Run the commands again and compare the addresses.

Result

The IPv6 addresses before and after reconnecting should be compared to determine whether any address changed. The actual result depends on the system's IPv6 privacy settings.

Conclusion

Temporary IPv6 addresses help improve privacy by reducing the long-term use of the same address for outgoing connections. Their behavior can be observed by comparing the IPv6 configuration before and after reconnecting.

Task 4 : Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

To create a Python script that stores restaurant names with their IPv6 addresses and searches for a restaurant using a partial name.

Program

```
restaurants = {
  "Zomato Cafe": "2001:db8:1::10",
  "Spice Garden": "2001:db8:1::20",
  "Pizza Hub": "2001:db8:1::30",
  "Food Palace": "2001:db8:1::40"
}
search = input("Enter restaurant name: ").lower()
found = False
for name, ipv6 in restaurants.items():
  if search in name.lower():
    print(name, "IPv6 Address:", ipv6)
    found = True

if not found:
  print("Restaurant not found.")
```

Sample Output

Enter restaurant name: pizza

Pizza Hub IPv6 Address: 2001:db8:1::30

Explanation

- A Python dictionary stores restaurant names and their IPv6 addresses.
- The user enters a full or partial restaurant name.

- The program searches for matching names without considering uppercase or lowercase letters.
- If a match is found, the restaurant name and its IPv6 address are displayed.
- If no match is found, the program displays Restaurant not found.

Result

The program successfully searches restaurant names using partial input and displays the matching IPv6 addresses.

Conclusion

This program demonstrates how Python dictionaries and string searching can be used to build a basic restaurant lookup system. The IPv6 addresses are sample documentation addresses, not real restaurant server addresses.

Task 5 : Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.